

Reese Ludwick

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Summary

Agentic AI engineer who **ships agentic systems to production and owns the agent loop end to end** — **planning, tool-calling, memory, and multi-step reasoning**. ~1 year building GenAI and multi-agent systems with LangGraph, LangChain, and Python: production MCP servers that give agents dynamic tool/function access, a multi-agent LangGraph system with a supervision agent, and agent-loop products — a live product (autoclip.studio) plus an in-progress agentic finance platform (CFO Agent) — not prototypes left on the bench. Comfortable scoping, designing, and owning agentic workloads cross-functionally under compressed timelines.

Experience

Software Associate — **Cognizant** (Remote, Dec 2025–Present) - Build GenAI and agentic AI systems with LangChain, LangGraph, and Python. - Design RAG pipelines and MCP server integrations that back **production** workflows. - Scope, prototype, and deliver AI data tools cross-functionally with client and internal teams.

Freelance AI Engineer & Full-Stack Developer — **WorldCovers / APMC** (Remote, May 2026–Present) - Ship production web properties and the WorldCovers platform to real clients on compressed timelines. - Designed a 14-table PostgreSQL schema; built Django REST + React/TypeScript surfaces. - AI-augmented engineering (Claude Code); direct stakeholder coordination and delivery ownership.

GenAI Developer Contractor — **Cognizant** (Remote, Jun–Dec 2025) - Built custom **FastMCP servers** and integrated external MCP services with AWS to give agents **dynamic tool/function access** at runtime. - Delivered GenAI POCs; ran prompt engineering and model evaluation. - Assembled modular, production-style agentic workflows.

IT Student Helper — **James Rumsey Technical Institute** (2019–2021) - Resolved 300+ support tickets; led a Wi-Fi overhaul cutting latency 25%; completed 100+ AV/network setups.

Projects

Agentic Software Development Team — *LangChain, LangGraph, Amazon Bedrock (Cognizant R&D)* - **Multi-agent LangGraph system** automating the full software build lifecycle end to end. - Specialized agents for code generation, testing, security review, and **supervision** — coordinated **planning loops** and **multi-step reasoning** across agents. - Backed by Amazon Bedrock plus OSS ML inference backends.

CFO Agent — *Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API* - FastAPI + LangGraph agent owning the full **agent loop**, with PostgreSQL persistence and Redis caching. - **Hybrid context architecture for agent memory** — short-term working state in Redis, durable state in PostgreSQL. - Financial data via CSV/OFX import and an encrypted vault; multi-tenant-ready.

CreatorClip / autoclip.studio — *Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg, YouTube OAuth* - Live production SaaS: Stripe billing; CI auto-deploys to prod on push to main. - Async Celery: WhisperX transcription, Voyage + pgvector embeddings, ffmpeg rendering; creator-“DNA” module; YouTube OAuth.

Customer Call Data Analysis & Insights Engine — *Python, NLP, Sentence Transformers (Cognizant R&D)* - NLP clustering of calls into semantic groups; agentic AI for call consolidation and executive reporting.

Technical Skills

Agentic / LLM: LangGraph (multi-agent, planning loops, supervision), LangChain, MCP servers (tool/function calling), RAG, agent memory (Redis + PostgreSQL + pgvector), Amazon Bedrock, Anthropic API, prompt engineering, model evaluation **Languages / Data:** Python, SQL, PostgreSQL, Redis, pgvector, Pandas, NumPy **ML:** Sentence Transformers, TensorFlow **Frameworks / Infra:** FastAPI, Celery, Django REST, React/TypeScript, Docker **Tooling / Cloud:** Linux/Ubuntu, Git, GitHub, AWS, Azure Fundamentals, VS Code, Claude Code, Jupyter

Education & Certifications

BS, Computer Science — **Shepherd University** (Aug 2021–May 2025), Major GPA 3.43 **Certifications:** TensorFlow Developer Specialization; Linux + Git (Boot.dev); AZ-900; GitHub Copilot; Generative AI Context Engineering (Cognizant)